

System Architecture

Manim AI Video Generator & Canvas Annotator

The project is an AI platform that converts PDF documents and prompts into 3Blue1Brown-style mathematical and conceptual explainer videos, complete with an interactive canvas for visual Q&A annotations.

System Overview & Workflow

The pipeline below shows the end-to-end flow, from a user's PDF and prompt through to a rendered explainer video:

Step	Flows Into
User PDF + Prompt	Next.js Frontend
Next.js Frontend	Backend API / Modal Cloud
Backend API / Modal Cloud	DocumentEmbedderAgent
DocumentEmbedderAgent	Qdrant Vector DB (via gemini-embedding-2)
DocumentEmbedderAgent	StoryAgent
StoryAgent (gemini-2.0-flash)	ValidatorAgent
ValidatorAgent	CodeGenAgent (Approved Script)
CodeGenAgent	CIPipelineHarness (Manim CE Code)
CIPipelineHarness	Code Valid? -> No: back to CodeGenAgent Yes: continue
CIPipelineHarness (Valid)	RendererAgent
RendererAgent (Manim CLI + ffmpeg)	UploaderAgent
UploaderAgent (DigitalOcean / Local)	MP4 Explainer Video

Key Architecture Components

1. Multi-Agent Pipeline (backend/pipeline/)

Built with LangGraph 1.2 StateGraph, orchestrating AI agents across a shared VideoJob state object:

- **DocumentEmbedderAgent:** Reads PDF documents via pypdf, checks constraints (max 20 pages), chunks text into ~500-token segments, and embeds them into Qdrant.
- **StoryAgent:** Conducts semantic RAG search over Qdrant using gemini-2.0-flash to structure a multi-scene narrative script with visual layouts and LaTeX formulas.
- **ValidatorAgent:** Validates the script layout and triggers revision loops if content is missing.
- **CodeGenAgent:** Converts the narrative script into clean Python code defining a MainScene(Scene) class targeting Manim CE (v0.20.1).
- **CIPipelineHarness:** Performs dry-run code verification (AST parsing + py_compile). On syntax or structural errors, it feeds compiler error traces back to CodeGenAgent for automated retries (up to 3 attempts).
- **RendererAgent:** Executes manim render CLI to output high-resolution .mp4 video files.

- **UploaderAgent:** Uploads completed video assets to DigitalOcean Spaces S3 storage via boto3==1.35.99.

2. RAG & Vector Storage (backend/rag/qdrant_store.py)

- **QdrantRAGStore:** Connects to your Qdrant Cloud Cluster (manim-docs collection).
- **GeminiEmbeddings:** Uses Google's gemini-embedding-2 (3072-dimensional vector space) with cosine similarity matching.
- **Payload Filtering:** Restricts searches strictly to the active job_id so PDF context is strictly isolated.

3. Interactive Canvas Annotation & Stitching (backend/pipeline/annotation_handler.py)

Allows users to pause the video at any frame, draw/highlight on screen, add a question, and receive an updated video:

- **Gemini Vision:** Reads the base64 composite frame image and user drawing.
- **Context Retrieval:** Queries Qdrant for PDF document context; falls back to Tavily Web Search (ResearchAgent) for out-of-document questions.
- **Clip Generation:** CodeGenAgent generates a dedicated AnnotationScene Manim clip answering the highlight.
- **ffmpeg Stream-Copy Stitching:** VideoStitcher inserts the rendered explanation clip into the original video at the exact pause timestamp without re-encoding existing video segments.

4. Dual Execution Backend

- **Modal Cloud (backend/modal_app.py):** Uses @modal.fastapi_endpoint() on an A10G GPU. Automatically provisions a Debian Linux container with pre-configured manim==0.20.1, ffmpeg, and texlive-full LaTeX engine.
- **Local Server (backend/local_server.py):** Fast local FastAPI server using python-dotenv for instant local testing and debugging.

5. Next.js 15 Frontend (frontend/)

- **Design System:** Dark theme, glassmorphism, glowing accents, Google Inter/Outfit typography.
- **Components:**
 - **UploadForm.tsx:** Drag-and-drop PDF uploader + prompt input.
 - **VideoPlayer.tsx:** Custom HTML5 player with frame-exact time tracking.
 - **AnnotationCanvas.tsx:** Transparent freehand drawing canvas + frame capture composite engine.
 - **CommentBox.tsx & TimelineScrubber.tsx:** Annotation controls.
- **API Proxy Routes:** Proxies requests from Next.js to backend Modal/Local endpoints, handling base64 conversion and job polling.

Stack & Dependencies

Category	Details
Language & Frameworks	Python 3.11, Next.js 15 (React 19), FastAPI, Uvicorn
AI Models	Google Gemini 2.0 Flash, Gemini Embedding 2, Tavily Web Search
Animation & Video Engine	Manim Community Edition v0.20.1, ffmpeg, TeX Live (LaTeX)
Agent Graph	LangGraph 1.2, LangChain 1.0
Vector Database	Qdrant Cloud (3072-dim Cosine Distance)
Cloud Infrastructure	Modal Cloud (A10G GPU), DigitalOcean Spaces (S3)